

5. The differential (a kind of gear mechanism) controls the turning of the wheels.

4. The driveshaft under the car transmits power to the rear axle.

3. Combinations of gears in the gearbox increase the car's speed and force.

2. The rotating crankshaft transmits power to the gearbox.

Power transmission

Today's motor vehicles have a transmission system that transmits power from the internal combustion engine (see p.148), through the gearbox, to the wheels.

The gearbox controls the speed of the car by adjusting the speed ratios between the engine and the wheels.

1. The moving pistons turn the crankshaft, which converts up-and-down motion of the pistons to rotary motion.

Horsepower

Traditionally, the power of an engine was measured in horsepower (hp). The term was coined by Scottish engineer James Watt to help sell his steam engines—he boasted that one of his engines could harness the power of 200 horses at once. One hp is equivalent to 746 watts, the standard unit of power, named after James Watt.



This automobile from 1897 had a 10-hp engine.

“I sell here, Sir, what all the world desires to have—POWER.”

Matthew Boulton, James Watt's business partner, 1776



A steamboat on the Mississippi River, c 1859

The electric charging station connects the car safely to the national grid.



A driver recharges her electric car

Steamships

As well as powering locomotives, steam engines were soon being used on ships. Paddle steamers carrying cargo and passengers on major rivers, such as the Mississippi, opened up new areas of the US to agriculture and trade in the 19th century. The steam from the boiler turns the paddlewheels, driving the steamer forward.

Greener engines

Gasoline and diesel engines burn a lot of fuel and give off heavily polluting exhaust gases. Today, hybrid cars, which combine a gasoline engine and an electric motor, and all-electric cars are increasingly popular.

1876

German engineer Nikolaus Otto invented the gasoline-driven four-stroke internal combustion engine. It soon led to the development of the automobile.

1897

Rudolf Diesel, a German inventor, built the first diesel engine. Diesel engines produce more power than gasoline engines and can be used to pull heavier loads.

1926

American engineer Robert Goddard invented the liquid-fuel rocket engine. Scarcely recognized at the time, he is now regarded as the father of modern rocketry.

1937

English engineer Frank Whittle and German engineer Hans von Ohain independently developed the turbojet engine that increased the speed of aircraft.

Model of Robert Goddard's rocket

